

Sai Srikar Vechalapu

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Summary

Data Engineer with 5+ years building data lake infrastructure, streaming pipelines, and dimensional warehouse models on AWS. Own a layered S3 data lake and the streaming path behind it, processing 12M+ daily supply-chain events into a multi-billion-row Redshift warehouse. Migrated a 50-workflow legacy ETL estate to Airflow and added data quality checks gating 200+ production tables.

TECHNICAL SKILLS

Languages: Python, SQL, PySpark, Bash

Data Platforms: Apache Spark (Batch and Structured Streaming), Kafka, Airflow, dbt, Databricks

AWS: S3, Redshift, Glue, Athena, Step Functions, EventBridge, CloudWatch, IAM

Data Warehousing: Dimensional modeling, ELT, CDC, SCD Type 2, partitioning, incremental loads, data quality frameworks

Databases: PostgreSQL, MySQL, Oracle, Snowflake, DynamoDB

DevOps & Reporting: GitHub, CI/CD, Docker, Tableau

Experience

McKesson

Aug 2024 – Present

Data Engineer

- Own the S3 data lake behind supply-chain analytics, processing high-volume fact data in Databricks and PySpark with partitioning, schema validation, and incremental loads before publishing curated Parquet for Athena and downstream reporting. Compacted small files into 128 MB Parquet and applied partition pruning to cut unnecessary S3 scans and speed up queries.
- Operate the near-real-time ingestion path for order, pricing, and inventory events, running Kafka into Spark Structured Streaming into S3 and Redshift. Handles 12M+ events per day at a 6K per second peak, with registry-backed schema validation, check-pointed exactly-once writes, dead-letter replay, and a 15-minute freshness SLA alarmed in CloudWatch.
- Migrated 50+ Informatica workflows to Airflow, breaking monolithic mappings into 180+ restartable tasks with explicit dependencies and SLA sensors. Removed a daily rerun queue averaging 6 manual fixes per week, and analysts now run 24-month backfills through parameterized DAGs.
- Rebuilt the pricing and supply-chain marts in Redshift with conformed product, customer, and location dimensions, SCD Type 2 on contract pricing, and CDC watermarked incremental loads against a 5B-row order-line fact. Retired 9 overlapping team-owned extracts in favor of one certified model.
- Cut P95 runtime on the top-20 reports from 38 to 5 minutes by fixing distribution and sort keys on the two largest fact tables, replacing repeated join patterns with materialized views, and eliminating a full scan from a non-sargable date filter.

Uber

May 2021 – May 2022

Data Engineer

- Developed PySpark pipelines processing 3M+ daily trip events for forecasting and pricing analytics, using watermarked windows for late-arriving events and partition-overwrite writes across date and city partitions on S3.
- Onboarded 15+ source systems into the S3 data lake and Redshift through AWS Glue, standardizing a landing-to-curated layout with partition-aware crawlers to keep downstream scan costs predictable.
- Implemented 40+ dbt models with tests and documentation for marketplace demand, driver supply, and retention datasets, replacing duplicated ad-hoc SQL across marketing, sales, and operations with tested, lineage-tracked models.
- Converted recurring one-off analyst SQL requests into governed Tableau data sources, removing roughly 10 hours per week of manual pull work from the team.

Cipla

Nov 2018 – Apr 2021

Data Engineer

- Integrated SAP, Salesforce, and legacy Oracle systems into a central warehouse using Talend, serving supply-chain and operational reporting for 8 departments across 120+ mapped source tables.
- Designed dimensional models for manufacturing, inventory, and sales reporting, then replaced nightly full refreshes with CDC-driven incremental loads. Batch window dropped from 7 hours to 90 minutes, clearing reports before regional business open.

Projects

Data Pipelines with Airflow

GitHub Repo

- Built Airflow ETL pipeline for S3-to-Redshift warehouse loads using custom StageToRedshift, LoadFact, LoadDimension, and DataQuality operators; supported templated backfills, SQL transforms, and fact/dimension loads.

Cross-Platform Marketing Analytics

GitHub Repo

- Built Python/SQL pipeline standardizing Facebook, Google, and TikTok ad CSVs into one BigQuery model with 13 data-quality checks and 3 analytics views for self-service Looker Studio reporting.

Education

University of Memphis

Aug 2022 – May 2024

Master's, Management Information Systems